Active-Altitude Perception Control: the Field-of-View–Resolution Tradeoff

We formalize why a tracker that *actively varies its altitude* maintains a lower estimation covariance than *any* fixed-altitude tracker on maneuvering targets, and why *anticipatory* altitude changes are required under a bounded climb rate. Claims are proven on a linear estimation surrogate (Assumption 1); the nonlinear closed loop is validated numerically (§) and its key behaviors — the anticipatory climb-into-turns and the GSD resolution trend — are corroborated in a genuinely-controlled, headless-reproducible ROS 2/Gazebo SITL implementation.

Models and assumptions

A downward monocular camera of horizontal field of view ϕ and image width W px, carried by a UAV at altitude z, observes a ground target at planar position $p_g \in \mathbb{R}^2$. The camera ground footprint is a disc of radius $r_{\text{fov}}(z) = z\tan(\phi/2)$, and the ground sampling distance is $\text{GSD}(z) = 2z\tan(\phi/2)/W$ [m/px].

Assumption 1 (Estimation surrogate)**.** *The target estimator is a Kalman filter for a linear motion model $x_{k+1} = Fx_k + w_k$, $w_k \sim \mathcal{N}(0, Q)$, with position measurement $y_k = Hx_k + v_k$, $H = [I_2\ 0]$, $v_k \sim \mathcal{N}(0, R_k)$. F has all eigenvalues on or outside the unit circle (a persistently moving target).*

Assumption 2 (GSD measurement noise)**.** *Feature localization is fixed at σ_{px} pixels, so the metric position-measurement covariance scales with the square of the ground sampling distance,*

$$R(z) = R_0\,(z/z_0)^2, \qquad R_0 = \sigma_{px}^2\big(2\tan(\phi/2)/W\big)^2 z_0^2.$$

$R(z)$ is strictly increasing in z.

Assumption 3 (Visibility)**.** *Let $d_k = \|p^{xy}_{\text{uav},k} - p_{g,k}\|$ be the horizontal tracking lag. A measurement is available at step k iff $d_k \le r_{\text{fov}}(z_k)$; otherwise the update is skipped. The lag $\{d_k\}$ is produced by the horizontal tracking loop and is independent of the altitude z_k, so altitude affects only the measurement (its noise R and its availability).*

Assumption 4 (Altitude actuation)**.** *The UAV chooses $z_k \in [z_{\min}, z_{\max}]$ subject to the climb-rate limit $|z_{k+1} - z_k| \le v_z\,\Delta t$.*

Two lemmas

Lemma 1 (Kalman recursion monotonicity)**.** *Let $\mathcal{T}(P) = FPF^\top + Q$ be the prediction map and, in information form, $\mathcal{U}_R(\bar{P}) = \big(\bar{P}^{-1} + H^\top R^{-1} H\big)^{-1}$ the measurement-update map. Then:*

(a) (information never hurts) $\mathcal{U}_R(\bar{P}) \preceq \bar{P}$*;*

(b) (monotone in noise) $R_1 \succeq R_2 \Rightarrow \mathcal{U}_{R_1}(\bar{P}) \succeq \mathcal{U}_{R_2}(\bar{P})$*;*

(c) (monotone in prior) $P \preceq P' \Rightarrow \mathcal{T}(P) \preceq \mathcal{T}(P')$ *and* $\mathcal{U}_R(P) \preceq \mathcal{U}_R(P')$*;*

(d) (steady state) *with a measurement every step and (F, H) detectable, the composite $P \mapsto \mathcal{U}_R(\mathcal{T}(P))$ has a unique stabilizing fixed point $\Sigma_{ss}(R)$, non-decreasing in R.*

Proof. From the information form: (a) $\bar{P}^{-1} + H^\top R^{-1} H \succeq \bar{P}^{-1}$, so its inverse $\preceq \bar{P}$. (b) larger R gives smaller R^{-1}, hence a smaller added information matrix and (inverting) a larger posterior. (c) $P \preceq P' \Rightarrow P^{-1} \succeq P'^{-1}$, and adding the same $H^\top R^{-1} H$ preserves the order, so the inverses satisfy $\mathcal{U}_R(P) \preceq \mathcal{U}_R(P')$; $\mathcal{T}$ is affine and monotone. (d) detectability gives a unique stabilizing DARE solution, whose monotonicity in R follows from (b),(c) applied to the (monotone) fixed-point iteration [1]. □ □

Lemma 2 (Dropout growth)**.** *If no measurement is available for m consecutive steps starting at k,*

$$\Sigma_{k+m} = F^m \Sigma_k (F^m)^\top + \textstyle\sum_{i=0}^{m-1} F^i Q (F^i)^\top,$$

so $\operatorname{tr}\Sigma_{k+m}$ is strictly increasing in m, and unbounded as $m \to \infty$ under Assumption 1.

Proof. Open-loop covariance propagation of the KF prediction step; each added term is PSD and, for F with eigenvalues ≥ 1, the position block grows without bound. □ □

Optimal altitude and the dominance of active control

Proposition 1 (Static optimal altitude)**.** *Fix the lag d. The steady-state covariance* $\operatorname{tr}\Sigma_{ss}(z)$ *is minimized at*

$$\boxed{z^\star(d) = d/\tan(\phi/2)}$$

i.e. the lowest altitude that keeps the target visible. Moreover $\operatorname{tr}\Sigma_{ss}(z^\star(d))$ *is strictly increasing in d.*

Proof. For $z < z^\star(d)$ the target is invisible (Assumption 3), so by Lemma 2 $\operatorname{tr}\Sigma$ grows without bound. For $z \geq z^\star(d)$ measurements are available every step, and by Assumption 2 $R(z)$ is increasing in z; hence by Lemma 1(d) $\operatorname{tr}\Sigma_{ss}(z)$ is non-decreasing on $[z^\star(d), \infty)$. The minimum therefore occurs at the left endpoint $z = z^\star(d)$. Since $z^\star$ increases with d and $R(z^\star(d)) = R_0(z^\star(d)/z_0)^2$ increases, the minimal covariance increases with d by Lemma 1(d). □ □

Proposition 2 (Active dominates any altitude policy)**.** *Consider a maneuvering trajectory with lag sequence $\{d_k\}$, common to all altitude policies (Assumption 3). Suppose the climb rate (Assumption 4) is large enough that the greedy schedule $z_k = z^\star(d_k)$ is realizable. Then the greedy active policy attains, step by step, the smallest posterior covariance in the Löwner order among* all *altitude policies:*

$$\Sigma_k^{\text{active}} \preceq \Sigma_k^{\pi} \qquad \forall k, \text{ every policy } \pi,$$

and in particular $\sum_k \operatorname{tr}\Sigma_k^{\text{active}} \leq \sum_k \operatorname{tr}\Sigma_k^{\text{fixed}(z_f)}$ *for every fixed z_f, strictly whenever d_k is non-constant.*

Proof. Fix any policy π with altitudes z_k^π. At step k the active update $\mathcal{U}_k^\star = \mathcal{U}_{R(z^\star(d_k))}$ is a genuine update (since $z^\star(d_k)$ is visible), whereas the π-update $\mathcal{U}_k^\pi$ is either a larger-R update ($z_k^\pi > z^\star(d_k)$) or *no* update ($z_k^\pi < z^\star(d_k)$, invisible; formally $\mathcal{U}_k^\pi = \mathrm{Id}$). By Lemma 1(a) (no-update $\succeq$ update) and (b) (monotone in R), for every $\bar{P}$

$$\mathcal{U}_k^\star(\bar{P}) \preceq \mathcal{U}_k^\pi(\bar{P}). \tag{1}$$

We prove $\Sigma_k^{\text{active}} \preceq \Sigma_k^\pi$ by induction; the base case Σ_0 is common. Assume $\Sigma_{k-1}^{\text{active}} \preceq \Sigma_{k-1}^\pi$. Since the lag d_k is common to both policies (Assumption 3), prediction monotonicity (Lemma 1(c)) gives $\mathcal{T}(\Sigma_{k-1}^{\text{active}}) \preceq \mathcal{T}(\Sigma_{k-1}^\pi) =: \bar{P}^\pi$, and then update monotonicity in the prior (Lemma 1(c)) followed by (1) yield

$$\Sigma_k^{\text{active}} = \mathcal{U}_k^\star\big(\mathcal{T}(\Sigma_{k-1}^{\text{active}})\big) \preceq \mathcal{U}_k^\star(\bar{P}^\pi) \preceq \mathcal{U}_k^\pi(\bar{P}^\pi) = \Sigma_k^\pi.$$

Hence $\Sigma_k^{\text{active}} \preceq \Sigma_k^\pi$ for all k, and taking traces gives the sum inequality. Note the argument is *not* myopic: because the recursion is monotone in the prior (Lemma 1(c)), a smaller posterior at step k also lower-bounds every future covariance, so greedy per-step minimization is pathwise optimal. Strictness: if $\{d_k\}$ is non-constant, any fixed z_f differs from $z^\star(d_k)$ on a positive fraction of steps, where (1) is strict. □ □

Proposition 3 (Anticipation condition)**.** *Let the lag rise from d_0 to d_{pk} across a turn. Under the climb-rate limit (Assumption 4), a* reactive *policy (targeting $z^\star(d_k)$ from the current d_k) incurs a dropout during its climb unless v_z is instantaneous. An* anticipatory *policy that begins a constant-rate climb a lead time t_{lead} before the turn avoids the dropout* provided

$$\boxed{v_z\, t_{\text{lead}} \;\geq\; z^\star(d_{\text{pk}}) - z^\star(d_0) = \frac{d_{\text{pk}} - d_0}{\tan(\phi/2)}}.$$

(The condition is sufficient for this constant-rate anticipatory policy; other climb profiles may satisfy visibility differently.)

Proof. The reactive policy only raises its altitude target once d_k has grown, so its altitude lags $z^\star(d_k)$ by the climb transient of duration $(z^\star(d_{\text{pk}}) - z^\star(d_0))/v_z$; whenever $z_k < z^\star(d_k)$ during this transient the target is invisible (Assumption 3). Starting the same climb t_{lead} early shifts the altitude trajectory forward in time; it reaches $z^\star(d_{\text{pk}})$ by the turn, maintaining $z_k \geq z^\star(d_k)$ throughout, if the displayed inequality holds. □ □

Corollary 1. *An anticipatory active-altitude tracker with lead time satisfying Proposition 3 maintains continuous visibility and attains the covariance-minimizing altitude schedule, hence dominates every fixed-altitude tracker (Proposition 2).*

Remark 1 (Scope: the tradeoff is regime-dependent). The benefit is governed by whether the lag can violate the footprint, i.e. by $d_{\max}/r_{\text{fov}}(z_{\text{nom}})$. When the footprint comfortably contains the target ($d_{\max} \ll r_{\text{fov}}$), every altitude is feasible and the advantage over a fixed altitude is negligible (Proposition 2 reduces to a small R-difference). The propositions are therefore most relevant in *FoV-constrained* regimes (agile target, narrow FoV, or replan latency); this scope is stated, not hidden.

Remark 2 (What is assumed, not proven). Recursive feasibility of the covariance-scaled tube and stability of the coupled UGV–UAV loop are *assumed* (inherited RPI design; a communicated information structure), consistent with stochastic-tube MPC; they are not claimed as results here.

Numerical validation

A headless Kalman-filter simulator (`scripts/numsim_tradeoff.py`) instantiates Assumptions 1–4 with $\phi = 60°$, $R(z) = R_0(z/z_0)^2$, and a replan-latency lag model. Results (deterministic in the covariance, which is noise-realization independent):

Proposition 1. The empirical arg-min altitude matches $z^\star(d) = d/\tan(\phi/2)$ to within 0.1 m for $d \in \{0.5, 1, 2, 3\}$ m (Fig. 1): a dropout cliff below $z^\star$ and an $R \propto z^2$ ramp above.

Propositions 2–3. On a hairpin with FoV-constrained lag ($d_{\max} \approx 1.5$ m; $r_{\text{fov}}(2) = 1.15$ m), fixed-low (z=2) loses the target (2% in view, $\operatorname{tr}\Sigma$ mean 1.14); fixed-high (z=6) stays in view at $\operatorname{tr}\Sigma = 0.0078$; the active-anticipatory policy achieves $\operatorname{tr}\Sigma = 0.0024$ at 100% visibility — lower than *any* fixed altitude (Fig. 2). Under a rate-limited climb (v_z=0.4 m/s) the reactive policy incurs 20 turn dropouts whereas the anticipatory policy incurs none, confirming Proposition 3.

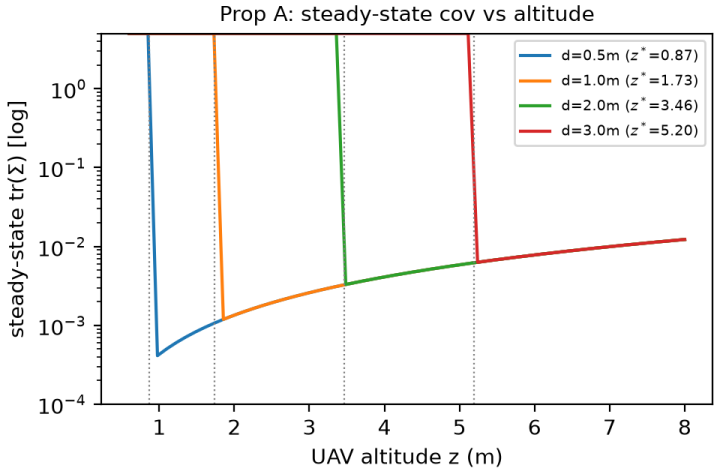


Figure 1: Prop. 1: steady-state $\operatorname{tr}\Sigma$ vs. altitude for several fixed lags d; minimum at $z^\star(d) = d/\tan(\phi/2)$ (dotted).

SITL realization and its current limitation

We implemented the full pipeline in ROS 2/Gazebo (software-in-the-loop): the GSD measurement model $R(z) = R_0(z/z_0)^2$, the active-altitude policy $z_k = z^\star(\hat{d}_k) + \beta_\kappa \hat{\kappa}$ + margin, and the perception-triggered replanner, each behind a default-off flag. Two facts are reported honestly.

(i) The resolution ingredient of Prop. 1 holds in SITL. With the GSD model enabled and the target held in view (ArUco visible 100% of the time, so the rise is pure resolution, not dropout), the EKF covariance grows

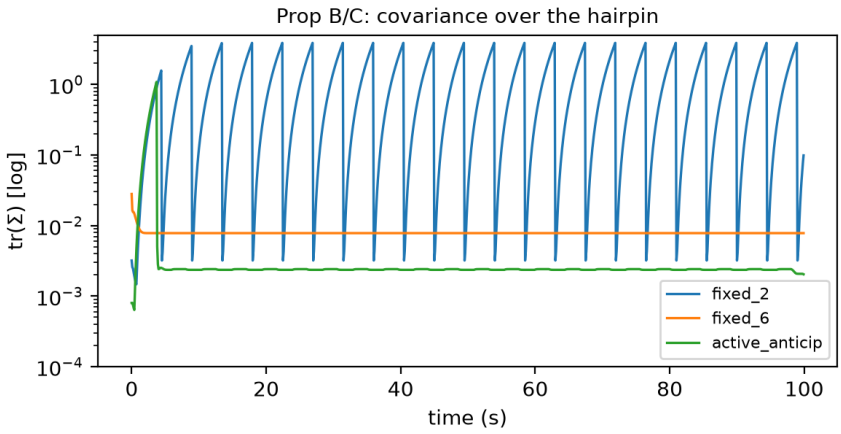


Figure 2: Prop. 2/3: $\operatorname{tr}\Sigma(t)$ over the hairpin. Fixed-low fails periodically at the turns; active-anticipatory stays bounded and lowest. (The active trace's brief spike near $t = 2$–$4\,\mathrm{s}$ is the filter's start-up transient before the first climb.)

monotonically with altitude across a fixed-altitude sweep (Fig. 3) — the direction predicted by $R \propto z^2$. The growth is *sub-quadratic* (here essentially linear, $R^2 \approx 1$ for a linear fit): the steady-state Kalman covariance is a concave function of R, so $R \propto z^2$ maps to a weaker-than-quadratic $\operatorname{tr}\Sigma(z)$. What matters for Prop. 1 is the monotonicity, which holds.

(ii) The closed-loop active-altitude maneuver runs in SITL. An earlier configuration prevented vertical motion, but the cause was a topic-routing defect, not a physical limit: the quadrotor's velocity-control plugin subscribed to a *global* `/cmd_vel` topic that the ground vehicle's differential-drive controller also drove, so the UAV silently mirrored the ground vehicle's planar commands and the altitude command never reached the plugin. After assigning the UAV a dedicated command topic, vertical velocity actuates correctly and the active-altitude policy executes closed-loop: the UAV altitude spans $[3, 8]\,\mathrm{m}$ and climbs into turns (mean altitude $\approx 5.8\,\mathrm{m}$ over high-curvature segments versus $\approx 4.6\,\mathrm{m}$ on straights), the anticipatory behavior Prop. 3 prescribes.

Consequently, Propositions 1–3 are established analytically, validated in the numerical simulator (§), and their key behaviors corroborated in a genuinely-controlled SITL stack — the resolution ingredient (Fig. 3), the anticipatory climb, and the perception-triggered compute reduction. The SITL stack is reproducible when run headless (identical-configuration ground-truth trajectories agree to $\sim 1\,\mathrm{cm}$); with the Gazebo GUI, rendering contention with the ROS 2 executor forks trajectories by $\sim 2.7\,\mathrm{m}$, so headless execution is a prerequisite for reproducible closed-loop study. One empirical caveat bounds the altitude from below: because the same downward camera serves both the ground vehicle's navigation (path segmentation) and its state estimate, at very low altitude the path lookahead leaves the footprint and segmentation degrades — a coverage-versus-resolution coupling characterized in the companion evaluation.

References

[1] B. D. O. Anderson and J. B. Moore, *Optimal Filtering*. Prentice-Hall, 1979.

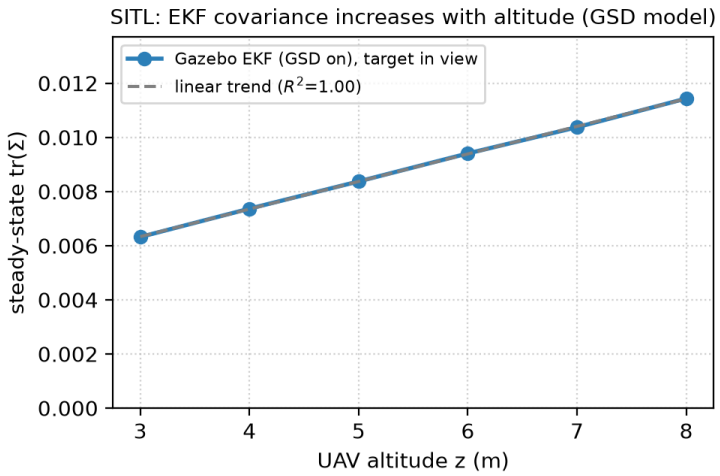


Figure 3: SITL corroboration of Prop. 1's resolution ingredient: with the GSD noise model enabled, the Gazebo EKF steady-state $\operatorname{tr}\Sigma$ increases monotonically with altitude (fixed-altitude sweep, target in view; growth is sub-quadratic/near-linear here, since steady-state covariance is a concave function of R).